

MANDATE

Make AI architecture load-bearing.

Turn fragmented GenAI, analytics, data-platform, API, microservice, and automation demand into an enterprise architecture system that accelerates delivery while preserving scientific trust, privacy, security, reliability, cost discipline, and reusable learning.

Company moment and role pressures

Why now

A broad portfolio of instruments, software, laboratory services, diagnostics, and pharmaceutical services creates heterogeneous workflows that must connect without false uniformity.

Real job

Define blueprints, governance, reference patterns, reviews, and execution guidance that reduce one-off architecture and invisible risk.

Operating tensions

- Innovation velocity vs enterprise assurance.
- Local scientific context vs reusable standards.
- Platform optionality vs common control points.
- Automation ambition vs human accountability.
- Scale economics vs resilient operating ownership.

Decision architecture

Make it easier to decide what changes, what gets funded or stopped, what becomes standard, what remains a legitimate exception, what escalates, and which humans retain authority.

OPERATING MODEL

The Scientific Capability Lattice

A reference architecture is load-bearing only when five layers stay connected. Weakness in one layer changes the disposition of the whole capability.

- 01 Context and data authority

Sources, lineage, quality, access, semantics, retention, and domain ownership.
- 02 Models, agents, and intelligence

RAG, models, agents, orchestration, evaluation, grounding, abstention, and change control.
- 03 Workflow and integration

APIs, events, ERP/CRM, laboratory systems, RPA, exceptions, and human handoffs.
- 04 Trust, governance, and control

Privacy, security, compliance, human authority, threat review, and decision records.
- 05 Operations, economics, and learning

SLOs, incidents, support ownership, unit cost, adoption, outcome evidence, and reuse.

Disposition logic

Promote when evidence is complete. **Revise** when one condition is weak. **Hold** when risk or ownership makes scale irresponsible. **Retire** when the capability no longer earns its burden.

PROOF AND INTERVIEW POSTURE

How to test the thesis in the room

Proof map

Transferable evidence

Vape-Jet: AI-first robotics operations. DudeWorth: AI readiness, RAG, agents, evaluation, governance. Compunetics: high-reliability controls. Amazon: scaled operating mechanisms.

Objection, stated plainly

No formal Enterprise AI Architect title and no claim of owning Thermo Fisher's exact platform estate. Pair with specialists and prove judgment through one bounded reference pattern.

Executive questions

- Where is architecture friction slowing squads most?
- Which patterns are already trusted and should be protected?
- Where do scientific requirements legitimately override defaults?
- Who owns reliability, change, and incident response?
- What evidence promotes a local proof to enterprise reference?
- How are cost, adoption, and outcomes connected to architecture decisions?

First 120 days

Read the system -> choose one bounded proof -> run one reference pattern -> compound the learning into reusable assets, clear decision rights, and an evidence-based disposition cadence.

Change burden

Business and scientific owners name outcomes and authoritative context. Architecture and platform teams expose paved roads and failure boundaries. Governance actors shape design early. Squads capture exceptions, friction, and evidence so patterns can learn.